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# AI vision and machine learning for enhanced automation in food industry: A systematic review

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ABSTRACT

Artificial intelligence (AI) vision and machine learning (ML) have been finding industry-wide transformations over the decade, ranging from improved quality control, decreased equipment downtime, and optimized operations. According to this review study, AI and ML technologies can achieve over 99 % accuracy through automation, compared to older approaches, which can only achieve 80–90 % accuracy through manual assessment. Machine learning algorithms improve production efficiency and reliability by minimizing equipment downtime. In robots, AI vision enhances operations, enhancing accuracy and efficiency while cutting labor expenses by 30 %. Up to 95 % optimization is achieved in supply chain efficiency, logistics, and inventory management by using AI-driven analytics. Customization is limited using traditional approaches, but with the help of AI and ML, sales can be increased by 15 %. The present study highlights how important artificial intelligence is in enhancing robotics, supply chain optimization, food safety compliance, predictive maintenance, and customized production amidst high capital investments. This paper discusses AI-ML-based applications in the food industry, which includes a gist of recent developments in the research, patents, government, and private sectors across continents and also in India's context. However, proper AI deployment requires addressing enduring issues, including data protection, complicated integration, ethical issues, and workforce upskilling. Artificial intelligence presents prospects for supply chain integration, personalized nutrition, and sustainability initiatives, even with initial capital expenditures. In order to upscale AI into the food processing business in the future while combining ethical commitments with technological advancement, effective stakeholder collaboration will be essential.

## 1. Introduction

### 1.1. Background

Industry 4.0 represents a new era of industrial development characterized by the integration of manufacturing operations systems with information and communication technologies (ICT) (Dalenogare et al., 2018), particularly the Internet of Things (IoT) (Padhiary, Tikute, et al., 2024; Saha et al., 2023); Cyber-Physical Systems (CPS) (Dafflon et al., 2021); additive manufacturing and 3D printing technologies (Padhiary, Barbhuiya, et al., 2024; Padhiary & Roy, 2024; Pradhan et al., 2024); machine learning (ML), artificial intelligence (AI) vision and image processing (Padhiary et al., 2023; Padhiary & Saha, Kumar, et al., 2024); and so on. Food processing automation involves the incorporation of technology and machinery to streamline and enhance different phases of food production, packaging, and distribution. The process involves a range of activities, such as sorting, grading, packaging, and quality control. Its purpose is to improve efficiency, minimize human mistakes, and maintain product uniformity (Duong et al., 2020). The automation of food processing has undergone significant advancements, primarily due to the necessity of meeting increasing demands, enhancing productivity, and upholding strict quality standards within the food industry. The field of food processing automation has undergone a significant transformation, with the introduction of mechanized assembly lines, advanced robotics, and sensor-based technologies (Dadhaneeya et al., 2023; Vashishth et al., 2022). This transformation has had a profound impact on every aspect of food production.

The application of AI and ML in the food industry has rapidly

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progressed, revolutionizing traditional methods and fostering innovation across various domains. AI and ML technologies have transitioned from experimental phases to integral components in optimizing food processing. These technologies have evolved to analyze complex data sets, predict consumer behavior, and enhance production efficiency. In a study by Hassoun et al. (2023), the authors highlight the advancements in machine learning algorithms aiding in food quality assessment, illustrating the potential for AI to revolutionize quality control and safety measures. Similarly, Srinivas et al. (2023) emphasize AI's role in predictive maintenance within food manufacturing, reducing downtime, and improving overall operational efficiency. The integration of AI and ML in the food industry continues to expand, offering unprecedented capabilities to address challenges and drive future innovations. However, it is important to consider the potential drawbacks of relying too heavily on AI and machine learning in food production, such as the risk of job displacement for human workers and the potential for errors or biases in algorithmic decision-making (Galaz et al., 2021). Additionally, the cost of implementing and maintaining AI technology may be prohibitive for some smaller food manufacturers (Duong et al., 2020).

Artificial intelligence is the foundation of machine intelligence, with the goal of giving robots abilities similar to human cognition, including learning, reasoning, and problem-solving. AI utilizes algorithms to enable machines to process data, detect patterns, and autonomously make judgments or predictions without the need for explicit programming for each individual activity. This essential component is in line with the concepts of machine learning, where models acquire knowledge from data in order to enhance performance through iterative processes. Taye (2023) It highlights the iterative aspect of machine learning, emphasizing that it depends on data to train models for tasks, including prediction and decision-making. AI methods often utilize neural networks, which are architectures inspired by the human brain, to process information and do intricate tasks like picture or speech recognition. Basheer and Hajmeer (2000), Kufel et al. (2023) Emphasize the importance of neural networks in AI applications, explaining their function in facilitating complex calculations and learning processes. The primary objective of AI is to develop systems that possess the ability to comprehend natural language, identify patterns, and independently adjust to new knowledge. Despite the fact that there has been significant progress, ongoing research consistently improves algorithms and strategies to increase artificial intelligence's capabilities in a variety of fields, as numerous researchers have emphasized.

### 1.2. *Understanding machine learning algorithms*

Understanding machine learning algorithms requires grasping the various approaches that enable machines to independently learn from data, recognize patterns, and make predictions or judgments. Machine learning involves a range of methods, such as supervised learning (Abdella et al., 2020); which uses labeled data to make predictions; unsupervised learning, which identifies patterns in unlabeled data; and reinforcement learning, which learns through trial and error with feedback from an environment (B. Singh et al., 2022; Wiering & Van Otterlo, 2012). Decision trees, support vector machines (SVM), and neural networks (Golbayani et al., 2020; Tsai, 2008; Westreich et al., 2010) are all types of supervised learning that use datasets with labels to teach models how to solve classification or regression problems. Also, researchers like Allaoui et al. (2020); Hurtik et al. (2020); Tyagi et al. (2022) explain how unsupervised learning techniques, like clustering or dimensionality reduction, enable computers to find hidden patterns or structures in data without explicit labels. According to Singh et al. (2022), reinforcement learning algorithms acquire knowledge by interacting with an environment and optimizing rewards through a series of sequential decision-making processes. Gaining a comprehensive understanding of these varied methodologies is essential for effectively utilizing machine learning in a wide range of applications across different industries.

Table 1 demonstrates the increasing fascination and progress in the domain of machine learning. As academics explore the limits of these techniques, it is crucial for practitioners to be informed about the most recent advancements and optimal methods. By using the capabilities of machine learning, organizations can gain novel insights, enhance their decision-making procedures, and foster innovation within their specific domains.

### 1.3. *Aim of the study*

This study specifically examines the progress of AI vision and machine learning in the food processing sector. It highlights the significance of AI and ML in enhancing quality control, increasing production efficiency, predicting maintenance needs, and optimizing supply chain operations (Rai et al., 2021; Ramirez-Asis et al., 2022). The review further investigates the difficulties encountered by these technologies, possible constraints, and ethical implications. The paper critically reviews academic literature, industry papers, and case studies that focus on the use of AI vision and machine learning to automate food preparation. Additionally, it evaluates aspects of data protection, obstacles to integration, and ethical considerations. The paper's objective is to conduct a thorough analysis of the use of AI vision and machine learning in the food processing industry. It seeks to identify specific areas that necessitate additional investigation and advancement. It provides guidance for future efforts to use these technologies' potential to transform food production and improve industry norms.

## 2. Literature review methodology

### 2.1. *PRISMA protocol*

The study utilized the preferred reporting items for systematic reviews and meta-analysis (PRISMA) technique to guarantee a thorough and transparent examination of the literature; following the method of Singh et al. (2023). This systematic methodology enables the identification, selection, and synthesis of pertinent studies on the influence of AI and ML in the domain of food processing. This study seeks to conduct a meticulous and organized review of the existing research environment by adhering to the PRISMA principles. The systematic review approach will identify major topics and trends in the literature, offering valuable insights into the current research on AI and ML in food processing. In addition, this study attempts to minimize bias and ensure the reliability and validity of the findings by following the PRISMA criteria (Singh et al., 2023).

Scopus database is selected for its comprehensive coverage of peer-reviewed research publications and citations in various global journals within the realm of science, amidst multiple databases like Google Scholar, PubMed, and Web of Science (WoS). Fig. 1 provides a detailed analysis of AI-Ml-based applications through a comprehensive literature

**Table 1**
Comparison of current work and existing literature.

| Criteria | Current work | Existing Literature |
|---|---|---|
| Fundamental principles of AI and ML | ✓ | ✓ |
| Understandable by non-programmers | ✓ | × |
| Prominent AI and ML frameworks and their key features | ✓ | ✓ |
| Applications beyond predictive analytics across industries | ✓ | × |
| Significant Government and Non-government efforts worldwide for AI-ML adoption in different fields | ✓ | × |
| Overview of key challenges and future directions in AI-ML implementation in various industries | ✓ | × |

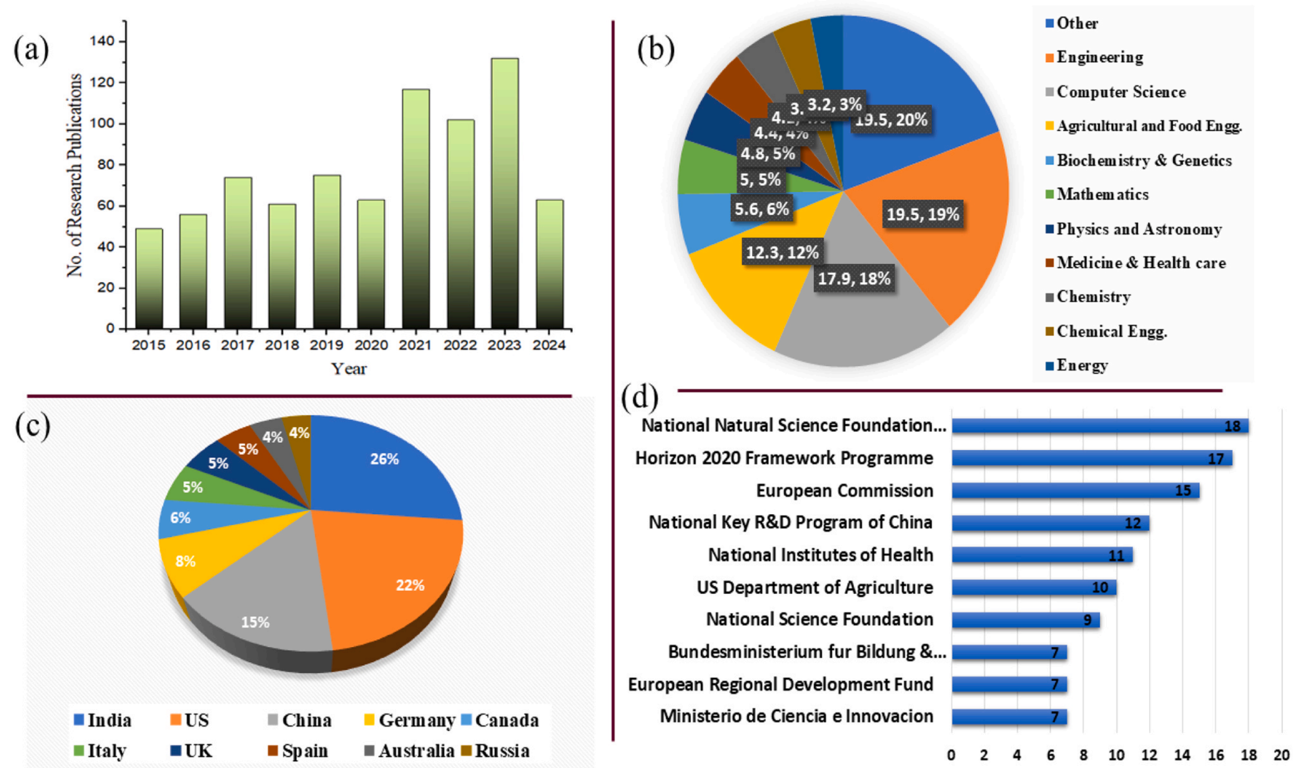


**Fig. 1.** Scholarly works on AI vision and ML worldwide in Scopus database(a) Number of research publications from 2015 to 2024; (b) Fields or domains of studies; (c) Countries active in studies; (d) Funding sponsored by major Departments/ Agencies (Data taken from Scopus Database on 23 June, 2024).

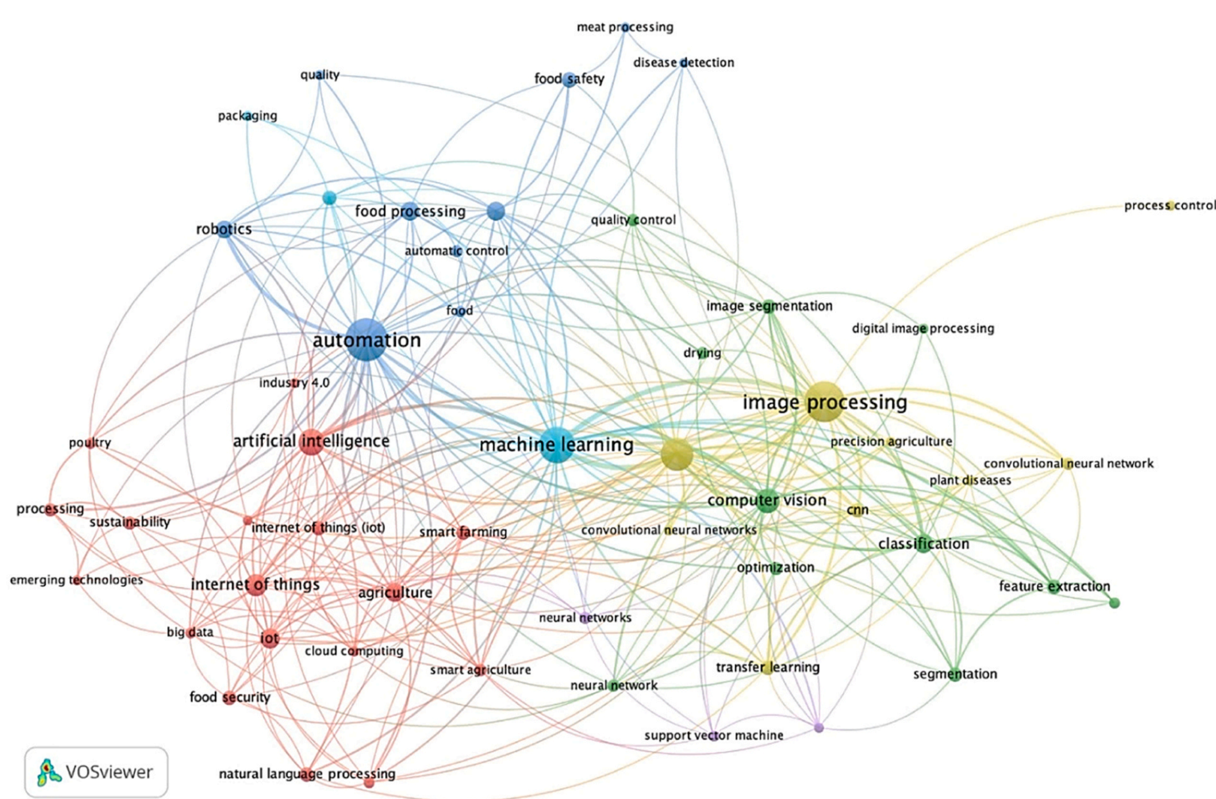


**Fig. 2.** The keyword network diagram in VOSviewer based on 55 most occurring keywords out of 2521, with a number of occurrences of a keyword being ≥ 5 (Based on 792 publications accessed from Scopus database with search keywords 'Automation in food processing, Food Processing, AI in Food Processing, ML in Food Processing' (23rd June 2024).

survey with data sourced from the Scopus database. There is an overall increasing trend in the number of publications in this domain, especially over the last 10 years (2015–2024) (Fig. 1**a)**. This signifies an escalating fascination with the prospective advantages of these technologies worldwide. Research on AI-ML-based applications is dominant in the disciplines of Engineering (19.5 %), followed by Computer Science (17.9 %) and Agricultural and Food Processing (12.3 %) (Fig. 1**b)**. The survey further shows that active scholarly works are being reported by researchers from India (26 %); followed by US (22 %) and China (15 %) across the globe (Fig. 1**c)**. Fig. 1**d** illustrates a horizontal bar graph depicting the 10 major funding agencies and Scientific and Research Departments across various countries; wherein the National Natural Science Foundation of China stood to be the most promising contributor in this regard.

### *2.2. Bibliometric analysis for the Interlinkage of keywords using Vos Viewer*

A bibliometric analysis is undertaken using Vos Viewer to examine the trends and interconnections of keywords in the literature survey. This analysis aims to visually depict the interconnections among fundamental concepts and issues in the domain of AI and ML as they relate to food processing. A bibliometric analysis was performed in VosViewer software following the methodology of Padhiary, and Saha, Kumar, et al. (2024). A keyword network diagram **(**Fig. 2**)** constructed based on 55 most occurring keywords out of 2521 from a total of 792 publications from the Scopus database with the search keywords 'Automation in food processing', 'Food Processing', 'AI in Food Processing', 'ML in Food Processing'. The diagram showed 6 clusters, indicating that AI/ML-based automation is a state-of-the-art technology and is a prime thrust area for research exploration in Food Processing. In the network, the keywords 'Automation' and 'Image Processing' seemed to be occupying the largest nodes and are well connected implying the rising interest of researchers and stakeholders in deploying these technologies in the market and business. In contrast, the keyword 'Process Control' was found to be very poorly connected with other keywords, raising concerns over the challenges faced in Process control for AI/ML-based technology implementations. This knowledge will be pivotal in directing future study endeavors and influencing policy determinations in the food business. Several researchers like Kalimuthu et al. (2024), El Jarroudi et al. (2024), Gardezi et al. (2024), Onyijen et al. (2024) suggested that the key challenges faced include ensuring robustness against variable conditions, integrating complex data sources for accurate decision-making, and overcoming regulatory and operational hurdles for effective real-world implementation in industries. However, to maintain focus and relevance, only the most influential research articles were included in the discussion, ensuring up-to-date coverage following the PRISMA principles as schematically shown in Fig. 3.

## 3. The role of AI vision in food processing automation

### *3.1. Quality control and assurance*

AI Vision plays a crucial role in maintaining stringent standards of food quality and safety during the production and packaging processes. The system utilizes sophisticated algorithms, including convolutional neural networks (CNNs) and deep learning models, to evaluate food items for flaws, irregularities, or impurities (Liu et al., 2021). These systems perform real-time analysis of visual data, allowing for quick detection and categorization of abnormalities in products such as fruits, vegetables, packaged goods, or processed commodities (Van de Looverbosch et al., 2021). AI vision systems can rapidly identify variations in color, shape, texture, or size by comparing photos with predefined quality standards and highlighting elements that do not satisfy the set criteria (Ramirez-Asis et al., 2022; Ramirez-Paredes & Hernandez-Belmonte, 2020). This technology improves the precision and efficiency of quality evaluations while reducing human mistakes, guaranteeing consistent adherence to strict quality control procedures and regulatory standards in the food business.

AI Vision is essential in the automated sorting and grading of agricultural goods. Automated systems with image recognition capabilities are used to confirm accurate packing and labeling, thereby minimizing mistakes and guaranteeing adherence to regulatory standards. AI Vision systems have the capability to identify impurities or foreign objects in the production line, allowing for prompt removal or reduction of contaminated items (Mittal et al., 2019; Padhiary, Kyndiah, et al., 2024). In addition, AI Vision systems have the capability to continuously monitor manufacturing processes in real-time, enabling the detection of any deviations from regular operating procedures. This feature significantly strengthens food safety regulations.

### *3.2. Predictive maintenance in manufacturing*

AI Vision greatly enhances predictive maintenance in manufacturing by offering proactive analysis of equipment health and effectively minimizing unforeseen downtimes (Zonta et al., 2020). By combining AI-driven vision sensors and machine learning algorithms,

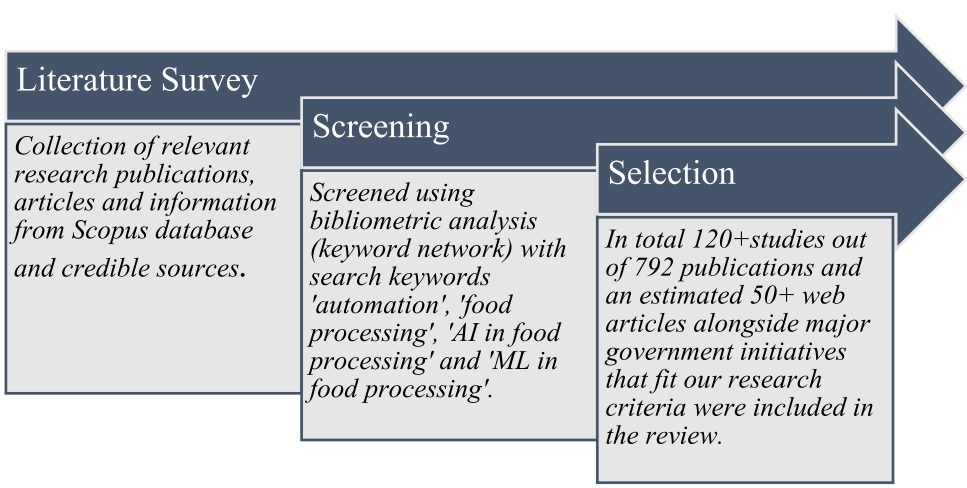


**Fig. 3.** Schematic of the PRISMA principles-based literature review methodology used for the study.

manufacturers can continuously monitor machinery and production lines in real-time. This allows them to identify even the slightest abnormalities or trends that may indicate possible breakdowns (Wang et al., 2023). These systems utilize advanced algorithms to examine visual data, such as fluctuations in temperature, vibrations, or deviations in structure, in order to predict the requirements for equipment maintenance (Serin et al., 2020). Predictive maintenance systems utilize previous data and ongoing observations to anticipate possible problems, enabling scheduled interventions and reducing unexpected downtime. This strategy enhances production operations, maximizes machinery uptime, and minimizes maintenance expenses by replacing components or performing maintenance tasks at the exact moment they are required (Afridi et al., 2022; Serin et al., 2020). Therefore, the implementation of AI vision-driven predictive maintenance in the food manufacturing business improves operational efficiency, extends the lifespan of equipment, and guarantees smooth production operations.

Predictive maintenance using AI-ML technologies has emerged as a crucial strategy in industrial settings to enhance operational efficiency and minimize downtime. For instance, ML models like Xtreme Gradient Boosting with oversampling techniques were successfully employed by Fernández-Peláez et al. (2024) to utilize large and imbalanced data from vibration sensors in the Monoblock machine. The model accurately predicted future stoppages with high accuracy within a 3-minute time window. Multiple ML algorithms like Random Forest (RF), Support Vector Machines (SVM), and Long Short-Term Memory (LSTM) were used by Srisuwan and Innet (2024) for predictive maintenance in high-pressure processing (HPP) system of food. They reported that the SVM model stood apart in the production line reducing the industrial maintenance costs by 5–10 % and positively impacting the product costs. However, proper adjustments in system configuration are essential before implementing the model in some other production situations or new production lines. Again, Oluleye and Ekun (2022) developed predictive maintenance models like Multiple Linear Regression (MLR) and Support Vector Regressor (SVR) for predicting the deterioration trends of a cassava pulveriser and evaluated their performance using the Root Mean Square Error (RMSE), Multiple R-squared scores, Mean Absolute Deviation (MAD), and Adjusted R-squared score metrics. The pulveriser's vibrational, sound, temperature, and feed capacity data were fed as the input variables to determine its time to the next failure occurrence and maintenance actions. The results validated the SVR model as the reliable one with more than 75 % accuracy in prediction. As per the study of Akhtar et al. (2024), a MATLAB-based linear regression model was found to detect anomalies, analyze trends, and forecast future maintenance requirements of cold stores in shrimp processing industries; with an accuracy of 95 %.

### 3.3. Robotics and automated processes

According to Hurd et al. (2005), AI Vision has transformed robotics and automated processes in food processing, enabling precise and adaptable automation. AI Vision is used with robotics to provide guidance and optimization for automated activities in food processing plants. Vision-guided robots, enhanced with artificial intelligence, employ advanced algorithms to interpret visual data in real-time. This enables them to successfully navigate intricate surroundings, recognize objects, and carry out tasks with remarkable precision and efficiency (Low et al., 2021). These systems have the capability to manipulate fragile food items, execute activities such as sorting, packaging, or assembly, and adjust to different conditions effortlessly. AI Vision technology enables robots to accurately see and interact with their environment, leading to increased efficiency, decreased errors, and consistent quality in food processing (Low et al., 2021; Sughashini et al., 2021). By utilizing AI Vision, robots, and automated processes, food processing operations can attain enhanced levels of accuracy, flexibility, and adaptability, leading to a transformative impact on the industry's capabilities and operating efficiencies.

### 3.4. Food supply chain (FSC) optimization and management

AI Vision plays a crucial role in FSC optimization by offering significant improvements and important insights across the supply chain (Baryannis et al., 2019). Through the utilization of AI-driven picture analysis, stakeholders can efficiently oversee and control several facets of the supply chain. AI Vision enables precise inventory management by visually monitoring stock levels, evaluating product quality throughout transportation, and improving warehouse operations (Baryannis et al., 2019; Min, 2010). Visual data analysis allows for quick detection of inconsistencies, possible product impairments, or abnormalities in packaging, ensuring quality control throughout distribution channels. In addition, AI Vision assists in enhancing the efficiency of logistics and distribution networks by examining live visual data to enhance the planning of routes, reduce transportation expenses, and guarantee punctual delivery (Dolz Ausina, 2023; Olugbade et al., 2022). By incorporating AI Vision technology into supply chain operations, stakeholders may optimize efficiency, minimize waste, and guarantee the reliable distribution of top-notch food items to consumers, thereby transforming the whole dynamics of the food industry's supply chain.

It has been demonstrated that in controlled-environment agriculture, AI models combined with IoT sensors optimize water and fertilizer supply, greatly minimizing waste and increasing crop yields (Padhiary, 2025). Furthermore, drone-based hyperspectral imaging has become a potent instrument for evaluating the quality of food, allowing for the early identification of contamination, ripeness, and disease in crops and food items. This enhances decision-making in the processing and agricultural industries (Kar & Dhal, 2025). Additionally, by anticipating changes in demand, reducing spoilage through clever cold-chain management, and enhancing traceability through blockchain technology, AI-driven supply chain optimization is revolutionizing food logistics (Dhal & Kar, 2024). These developments guarantee that food is delivered to consumers with greater safety standards and less waste. A more thorough understanding of AI's revolutionary role in guaranteeing global food security would result from the combination of these state-of-the-art studies, which would support conversations about automation, efficiency, and sustainability in contemporary agricultural systems.

### 3.5. Types of supply chain technologies

*iRFID tracking systems* play a crucial role in offering immediate visibility and traceability of products as they progress through the supply chain. RFID technology enables precise inventory control, mitigates the risk of stockouts, and enhances overall operational efficiency (Chen et al., 2007). By using this cutting-edge technology, organizations may optimize their supply chain processes and fulfill the increasing expectations of consumers in today's dynamic market.

*Inventory management software* is a critical technology that is transforming the way firms handle their supply chains. This program facilitates real-time tracking and monitoring of inventory levels for organizations, empowering them to make well-informed decisions regarding ordering and restocking (Arsan et al., 2013). Businesses can enhance their profitability by adopting inventory management software, which enables them to decrease surplus inventory, mitigate stockouts, and ultimately optimize their financial performance. Moreover, this technology has the capability to seamlessly merge with other supply chain systems, such as RFID monitoring, in order to offer a full perspective of the entire supply chain ecosystem (Atieh et al., 2016).

Thus, the integration of RFID tracking systems with inventory management software is facilitating firms in optimizing their operations and satisfying the requirements of the current dynamic market.

*Automated order processing systems* optimize inventory management by simplifying the procedures for receiving, completing, and delivering orders, thus increasing efficiency (Rudolph & Emmelmann, 2017). Through the process of automating these processes, organizations can effectively decrease the occurrence of human mistakes, decrease the amount of time required for processing, and enhance the overall level of customer satisfaction. Moreover, automated order processing systems can offer instantaneous updates on the status of orders, facilitating improved communication with clients and enabling more precise predictions of inventory requirements (Jaghbeer et al., 2020; Rudolph & Emmelmann, 2017). In today's fast-paced market environment, the integration of RFID tracking devices with automated order processing is fundamentally transforming the inventory management and customer order fulfillment practices of organizations.

These technologies collaborate to optimize the movement of goods from suppliers to manufacturers to retailers, ultimately enhancing overall operational efficiency and customer pleasure. By incorporating AI Vision technology, supply chain management can achieve heightened levels of accuracy and efficiency, resulting in substantial cost reductions and competitive advantages in the market. The future of supply chain management hinges on the seamless integration of innovative technologies such as AI Vision, which will facilitate a more streamlined and environmentally conscious food business.

### *3.6. Clinical significance of AI-based solutions*

AI-driven personalized nutrition is revolutionizing dietary interventions and chronic disease management by providing tailored recommendations based on an individual's genetic, metabolic, and lifestyle factors. By leveraging machine learning algorithms, AI can analyze vast datasets, including biomarker information, food preferences, and medical history, to develop precision-based dietary strategies. This approach enhances health outcomes by improving nutrient intake, reducing the risk of diet-related diseases, and optimizing metabolic health. For example, AI-powered platforms can assist in managing diabetes by predicting glucose fluctuations and suggesting real-time dietary adjustments, leading to improved glycemic control and reduced dependence on medication (Khalifa & Albadawy, 2024; Oka et al., 2019). Similarly, AI-driven nutritional interventions have shown promise in cardiovascular disease prevention by recommending heart-healthy diets tailored to an individual's lipid profile and genetic predisposition (Moz et al., 2023).

With the recent rise in AI-driven food analysis tools, such as ChatGPT-powered nutrition assistants, the intersection of GPT models and the food industry is becoming increasingly significant. These models can provide instant dietary advice, suggest meal plans based on real-time user inputs, and even analyze food labels for allergens and macronutrient composition. AI-driven chatbots are also being integrated into telehealth platforms, enhancing accessibility to expert dietary guidance and supporting behavioral change for weight management and lifestyle-related disorders (Naja et al., 2024; Prasad et al., 2025). Additionally, researchers like Oh et al. (2021); Ashton et al. (2023); Ponzo et al. (2024) claimed that AI-enhanced food tracking applications can effectively utilize image recognition to estimate portion sizes and nutritional content, improving dietary compliance and promoting healthier eating habits. While AI-driven personalized nutrition presents immense potential, challenges such as data privacy, algorithmic biases, and integration into clinical practice remain. However, ongoing advancements in AI, coupled with blockchain for secure health data management, can further enhance the credibility and applicability of AI-powered dietary interventions, paving the way for a new era of precision nutrition in healthcare.

## 4. Machine learning applications in food processing

### *4.1. Predictive maintenance in food safety and precision agriculture*

AI Vision has drastically altered the inventory management of the food business through the implementation of predictive analytics technologies. These systems utilize machine learning algorithms and visual data analysis to forecast inventory needs, anticipate swings in demand, and optimize stock levels. Their analysis involves examining patterns, seasonal trends, and customer habits in order to generate more accurate forecasts regarding future inventory requirements (Kharfan et al., 2021). This proactive approach to decision-making leads to a decrease in excess inventory and a decrease in instances where stock is depleted. Predictive analytics systems enhance the efficiency of supply chains and optimize storage capacities by synchronizing inventory levels with predicted demand. This improves the ability to quickly adapt and be efficient in the food sector. Inventory, consisting of raw materials, work-in-progress, and finished commodities, is essential in the supply chain (Vrat & Vrat, 2014). Raw materials inventory guarantees continuous production, work-in-progress inventory provides flexibility in modifying production plans, and finished goods inventory facilitates prompt and efficient fulfillment of client orders. Effectively overseeing all three categories of inventory enables organizations to optimize their supply chain operations and maintain a competitive edge in the market (Lin et al., 2020; Vrat & Vrat, 2014). For instance, a processing machinery manufacturer can promptly address unforeseen surges in demand for a particular model without having to wait for extra inventory. Efficiently controlling work-in-progress inventory enables the manufacturer to modify production schedules in order to give priority to popular models and reduce the number of unsold machines.

By increasing agricultural output, reducing losses, and guaranteeing effective resource use, AI-driven solutions are transforming food security. Real-time monitoring of agricultural diseases, weather patterns, and soil health is made possible by predictive analytics that use machine learning models. This allows for proactive actions to optimize yields and avert significant losses (Dhakshayani et al., 2024). Sustainable urban farming is a feasible way to address food shortages because smart agriculture technologies, like as hydroponics and vertical gardening, use AI-ML integrated IoT sensors to manage water, fertilizer, and light conditions (Dhal et al., 2024; Padhiary, 2025; Siregar et al., 2022). By preserving product integrity through robotic precision, AI-driven automation in harvesting, sorting, and packing minimizes post-harvest losses and lessens labor dependency (Dhal et al., 2024; Kar & Dhal, 2025).

Additionally, AI reduces supply chain bottlenecks while guaranteeing food safety and authenticity by improving food distribution efficiency using blockchain-based traceability and intelligent logistics (Dhal & Kar, 2025; Puthenveettil & Sappati, 2024). By tracking agricultural conditions, identifying irregularities, and maximizing the use of pesticides and fertilizers, AI-powered drones and self-driving cars support precision farming and increase production while minimizing environmental impact (Kar & Dhal, 2025; Padhiary & Kumar, 2025). Policymakers and business executives can create focused interventions to address food insecurity globally by incorporating AI into early warning systems for climate-related food shortages and using big data for demand forecasts. Increasing the scope of conversations about these AI-powered developments will offer a more comprehensive view of how technology is changing food availability and agricultural sustainability.

### 4.2. Smart packaging and labeling

AI Vision has influenced the food sector by converting conventional labeling procedures into more efficient and precise systems. AI-driven visual systems employ image recognition and machine learning algorithms to meticulously examine and authenticate packaging materials, labels, and codes on food items (Abid et al., 2024). Their purpose is to guarantee adherence to regulatory requirements and validate the accuracy of labeling information. In addition, they improve anti-counterfeiting efforts by identifying counterfeit packaging or labeling, thus protecting the integrity of the product and ensuring consumer confidence. AI vision systems play a role in improving traceability, reducing the chances of inaccurate labeling, maintaining product quality, and enhancing consumer safety and satisfaction (Liu et al., 2023).

Diverse forms of packaging, including vacuum-sealed pouches, aseptic containers, and modified environment packaging, are employed in the food business to prolong shelf life, preserve freshness, and safeguard against contamination (Ansari & Datta, 2003; Benyathiar et al., 2020; Mauer & Ozen, 2004). Food companies may enhance the security and compliance of their packaging, as well as increase consumer experiences and brand trust, by employing AI Vision technology. AI Vision technology can also aid in identifying possible problems or flaws in packaging prior to shipment, eliminating expensive recalls and safeguarding reputation, resulting in time and cost savings (Marucheck et al., 2011). The incorporation of AI Vision technology into packaging operations has the potential to transform the food sector by improving quality control, ensuring adherence to regulations, and providing superior products to consumers.

### 4.3. Personalization in food production

AI Vision has given a new horizon to the food manufacturing business through the provision of personalized solutions and customized experiences. By combining machine learning algorithms with visual data, these systems have the ability to adjust recipes, portion sizes, and ingredients to accommodate individual preferences or dietary needs (Boland et al., 2019). AI Vision aids in quality control by guaranteeing the precise assembly of tailored orders, validating ingredient accuracy, and tailoring packaging according to individual preferences. Such a degree of customization not only improves customer satisfaction but also enables food manufacturers to effectively cater to specialized markets and dietary limitations, changing the future of food production in a more consumer-focused manner (Zhou et al., 2019).

Artificial intelligence technology is employed to evaluate extensive quantities of data in order to comprehend client preferences and generate customized solutions that suit specific requirements. This not only optimizes the production process but also reduces waste and enhances efficiency in the supply chain. AI Vision technology uses cameras and sensors to thoroughly examine and evaluate each stage of the production process, starting from the acquisition of ingredients to the final packaging (Zhou et al., 2019). This technology guarantees that only products of the utmost quality are delivered to consumers. The high level of precision and accuracy not only enhances the overall quality of the end product but also ensures adherence to food safety standards and compliance with legislation. Moreover, AI Vision technology has the capability to identify potential impurities or alien substances in the manufacturing process, enabling prompt corrective measures to avert problems before they reach the end user.

Food businesses are increasingly using personalized packaging to improve consumer happiness and increase their market reach (Saguy & Sirotinskaya, 2014). Through the utilization of AI technology, organizations may evaluate consumer data and preferences to develop personalized packaging that deeply connects with their intended customer base (Lo et al., 2020). This fosters stronger brand loyalty and encourages customers to make repeat purchases, ultimately resulting in improved sales and profitability for the company. The possible future consequences of consumer-focused food production enabled by modern technology such as artificial intelligence are limitless. AI's capacity to evaluate extensive datasets and forecast consumer preferences empowers firms to foresee trends and provide inventive products that cater to individual tastes and dietary requirements (Lo et al., 2020). The transition to customized and eco-friendly food production has positive impacts on the environment and encourages healthier dietary choices among consumers.

## 5. Real-world implementations in food processing

### 5.1. Government initiatives

Various agencies and governments are exploring different use cases of this technology and are in different stages of implementation. The government's initiatives include financial support for research and development in sustainable food technologies, rules that encourage the adoption of AI-ML in food production, and incentives for enterprises that effectively minimize food waste and carbon emissions (Igeta & Nakamura, 2022; Zhou et al., 2022). These programs are designed to promote the adoption of ecologically sustainable and health-conscious techniques in food processing worldwide. Through the utilization of technology and the backing of government assistance, the outlook for food production in the future appears encouraging in terms of both sustainability and creativity. As consumers become more conscious of the environmental impact of their food choices, these initiatives will become more vital in ensuring a sustainable and healthy food supply for generations to come. Table 2a and Table 2b summarize the government initiatives and schemes launched in this regard in the global and the Indian context respectively.

### 5.2. Performance metrics and industry adoption

Evaluating the usefulness and acceptance of AI in the food processing business relies heavily on performance and industry adoption criteria. These measures evaluate the benefits of AI, including enhanced productivity, quality assurance, and cost efficiency. Business adoption indicators assess the level of incorporation and approval of AI in different industries, including major manufacturers, small and medium-sized enterprises (SMEs), and startups. Utilizing user input and analyzing case studies assists in defining areas that require enhancement. Companies like Bizom, Upliance, Intello labs, Spoonshot and so on are in the forefront of incorporating AI into food processing in India. This integration enhances supply chain management, food safety measures, and production efficiency as shown in Fig. 4. These companies exemplify how AI technologies are transforming the food processing industry (Table 3) and stimulating innovation in multiple areas of production with elevated performance metrics (Table 4).

### 5.3. Patents

Recent years have seen the patenting of several products and technologies in artificial intelligence for automation purposes, as well as machine learning algorithms designed to manage critical food processing operations. A concise overview of the most significant patents in this field is provided in this section **(**Table 5**)**. The patents cover the broad realm of artificial intelligence, machine learning, and robotics in transforming various aspects of the food industry, from processing and quality control to delivery and consumption optimization. Each invention aims to address specific challenges in food technology, starting from freshness detection to automated preparation and delivery, thereby advancing efficiency, quality, and customer satisfaction in food-related services.

**Table 2a**
Government initiatives in the global context (excluding India) [as of June 2024].

| Country | Objective(s) | Reference |
|---|---|---|
| United States | The 21st Century Cures Act (signed in Dec, 2016) aims to expedite medical treatment discovery, development, and delivery. The FDA and USDA are utilizing AI to detect contaminants and optimize supply chains. The US President committed to doubling nondefense AI R&D, and the General Service Administration launched the AI Center of Excellence. | (Shah et al., 2019; Zhou et al., 2022)https://trumpwhitehouse.archives.gov/ai/ |
| United Kingdom | The UK government has allocated £ 54 million to boost its AI and data science workforce, implementing the Agritech Strategy (launched in July, 2013) and National AI strategy (launched on 22nd Sept, 2021) to establish a regulatory framework for AI, aiming to promote growth, public trust, and global AI leadership. | https://www.gov.uk/government/news/54-million-boost-to-develop-secure-and-trustworthy-ai-research https://iuk.ktn-uk.org/events/ai-governance-key-uk-government-initiatives-and-future-direction/ |
| Malaysia | Malaysia's government is integrating technology into food safety and agriculture through the National Artificial Intelligence Initiative (in Dec, 2024) and the Food Safety and Quality Improvement Plan. The Agro Digital Innovation & Transformation Center is developing digital solutions for agriculture and food processing. Microsoft is investing $2.2 billion in Malaysia to create new cloud and artificial intelligence infrastructure. These initiatives aim to improve food safety, reduce waste, and boost industry competitiveness. | https://indiaai.gov.in/country/malaysia https://news.microsoft.com/apac/2024/05/02/microsoft-announces-us2-2-billion-investment-to-fuel-malaysias-cloud-and-ai-transformation/ |
| Canada | The Canadian government launched the Pan-Canadian Artificial Intelligence Strategy in 2017 to advance AI research, address ethical and regulatory issues, and establish the National Artificial Intelligence and Machine Learning Joint Program. Future plans include developing national AI standards, enhancing transparency, and collaborating with international partners. | https://ised-isde.canada.ca/site/ai-strategy/en |
| China | China's "Made in China 2025" plan released in 2015, led by the NDRC, focuses on AI applications in the food industry. The State Administration of Market Supervision and Administration has created a Food Safety Standardization Committee to improve AI quality control. The Chinese Academy of Sciences has developed an AI-enabled platform to detect foodborne risks. Key regulatory bodies include the National AI Governance Committee, the Cyberspace Administration of China, and the Ministry of Science and Technology. | https://digichina.stanford.edu/work/full-translation-chinas-new-generation-artificial-intelligence-development-plan-2017/ |
| Japan | The Japanese government put forward the "Society 5.0" concept in 2016, which combines AI, IoT, and big data to create an advanced social system. Key initiatives include the AI R&D Roadmap and AI Governance Framework and plans to establish a Digital Agency for responsible AI technology development. | https://www.sphericalinsights.com/blogs/japan-s-artificial-intelligence-ai-revolution (Igeta & Nakamura, 2022) https://www.fao.org/platform-food-loss-waste/news/news-detail/Japan-collects-and-shares-good-practices-to-accelerate-actions-for-food-loss-and-waste-reduction/en |
| Australia | The Australian government has implemented several key initiatives, including the 2017 AI Report, the 2020 AI and Data Strategy, and the 2020 Social Media and Online Safety Act. These initiatives aim to enhance AI adoption, combat misinformation, and develop AI-enabled sectors like healthcare, agriculture, and education. The Australian National Data Service and Data61 initiative also focus on interoperability and data sharing. Future Directions in Food Processing report highlights AI's potential for improving food safety, waste reduction, and efficiency. | https://www.finance.gov.au/sites/default/files/2024-06/National-framework-for-the-assurance-of-AI-in-government.pdf https://www.csiro.au/en/about/people/business-units/data61 https://data.gov.au/data/organization/australian-national-data-service |
| France | France's President has adopted the National Strategy for AI as part of the "France 2030" plan in Nov, 2021, aiming to expand AI talent and boost R&D for economic success. This follows previous initiatives promoting AI adoption, data protection, and skilled workforce development. AI is particularly beneficial in food processing for quality control, production optimization, and food safety. | https://digital-skills-jobs.europa.eu/en/actions/national-initiatives/national-strategies/france-national-strategy-ai |
| Netherlands | The Netherlands is prioritizing AI governance, focusing on ethical considerations, data protection, and transparency in the food processing industry. Key initiatives include the "AI for Everyone" program (in 2019), the "Data Governance Act," (in June, 2022), and the AI Coalition (in Oct, 2019), promoting responsible AI development and deployment. | https://ai-watch.ec.europa.eu/countries/netherlands/netherlands-ai-strategy-report_en#links https://www.government.nl/latest/news/2024/01/18/dutch-government-dutch-government-presents-vision-on-generative-ai |
| Germany | The government has enacted measures such as the "AI Strategy 2025" (in Nov, 2018) and "Digital Agenda 2025" (in 2016) to encourage the responsible development and implementation of AI. Additionally, they are undertaking the "AI in Agriculture" project to harness the potential of AI in the field of food processing. | (Keuter et al., 2021) |
| Egypt | The Egyptian government is employing AI-powered technologies, such as the "Smart Food" program (2023), to enhance food safety and quality. Additionally, they are utilizing the "Egyptian Food Bank" to streamline distribution processes and minimize wastage. The government intends to incorporate AI into every aspect of the food supply chain. | https://documents1.worldbank.org/curated/en/454961511210702794/pdf/Food-systems-for-an-urbanizing-world-knowledge-product.pdf |
| Ethiopia | The Ethiopian government is implementing the "Digital Ethiopia" program (launched in 2020) to boost food production and competitiveness and promote AI adoption through the "National AI Strategy" and "Digital Transformation of Agriculture" initiatives. | https://resilient.digital-africa.co/en/blog/2024/04/11/digital-ethiopia-2025-empowering-the-nation-in-the-digital-economy/ |
| Singapore | The "Food Factory of the Future" and "Agri-Food Tech" initiatives aim to improve food safety and quality through AI-powered monitoring and predictive maintenance since 2018. | https://www.dataprotectionreport.com/2024/02/singapore-proposes-governance-framework-for-generative-ai/ https://www.sfa.gov.sg/food-farming/singapore-food-story/singapore-food-story-r-d-grant-call |

(continued on next page)

**Table 2a** (*continued*)

| Country | Objective(s) | Reference |
|---|---|---|
| Italy | Italy is actively adopting Food 4.0, revolutionizing its economy by incorporating Industry 4.0 technology in 8.4 % of enterprises, and experiencing a 23 % growth in Agriculture 4.0 turnover since 2018. The nation is giving top priority to tackling the yearly expenses of 7 billion euros associated with food waste by implementing AI governance initiatives and making investments in food safety and efficiency. | https://impact.economist.com/perspectives/sites/default/files/food_4.0_report_final_2.pdf(Romanello & Veglio, 2022) |

## 6. Cost economics

The monetary consequences of integrating AI and machine learning (ML) into food manufacturing are diverse, including upfront capital expenditures, ongoing operating costs, and enduring financial advantages. This section explores the cost economics related to incorporating this modern technology into food processing workflows.

### *6.1. Initial investment costs*

Integrating AI and ML technologies into food production necessitates a substantial initial financial commitment. The expenditures include the acquisition and setup of sophisticated hardware, such as high-resolution cameras, sensors, and processing infrastructure, which are essential for AI vision systems (Lee et al., 2022). Furthermore, costs are accrued for the procurement of specialist software, the creation of customized algorithms, and the incorporation of these technologies into pre-existing production processes. Providing training to staff to proficiently run and maintain AI systems is a significant component of the initial investment (Babina et al., 2024).

### *6.2. Operational and maintenance costs*

The operational costs of AI and ML applications comprise the recurring expenses associated with operating, maintaining, and consistently enhancing AI systems. This encompasses routine software upgrades, upkeep of hardware, and expenses associated with either cloud services or on-site data centers for the purpose of storing and processing data (Javaid et al., 2022). In addition, there are expenses linked to recruiting and keeping proficient experts who can oversee AI-powered operations and analyze data findings.

### *6.3. Cost Savings and Efficiency Gains*

Although there are considerable upfront and ongoing expenses, the use of AI and ML applications can ultimately result in significant financial savings and improvements in efficiency. Improved quality control decreases waste and the necessity for rework, while predictive maintenance eliminates unforeseen equipment downtime, ensuring uninterrupted output (Wang et al., 2023; Zonta et al., 2020). The implementation of robots and artificial intelligence in automating repetitive operations leads to a decrease in labor expenses and an improvement in productivity. These enhanced efficiencies lead to increased profit margins and a faster return on investment.

### *6.4. Long-term financial benefits*

Over time, AI and ML have the potential to generate enduring economic advantages by streamlining supply chain operations, minimizing inefficiencies, and improving the quality of products. This, in turn, can result in increased consumer pleasure and stronger brand loyalty (Baryannis et al., 2019). Utilizing advanced analytics enables accurate anticipation of market trends and consumer preferences, leading to improved demand forecasting and inventory management, hence enhancing cost savings (Dolz Ausina, 2023). Furthermore, the implementation of customized nutrition and product options has the potential to provide additional sources of income, targeting certain market segments and ultimately expanding the entire market presence (Dolz Ausina, 2023).

### *6.5. Cost-benefit analysis*

Performing a comprehensive cost-benefit analysis is essential for food processing enterprises that are contemplating the adoption of AI and ML technologies. This entails assessing the overall cost of ownership (TCO) in comparison to the anticipated financial benefits and enhancements in efficiency (Saguy & Sirotinskaya, 2014). Companies should also take into account the scalability of AI solutions and the possibility of obtaining more benefits as AI technologies advance and enhance over time. Although the implementation of AI and ML in food processing incurs considerable upfront and continuous expenses, the potential for major long-term economic advantages justifies it as a strategic investment. For organizations striving to stay competitive and inventive in the changing food sector, it is crucial to carefully consider the relationship between these expenses and the anticipated return on investment.

The study that compared the performance and cost-effectiveness of traditional food processing methods to those that use AI and ML technologies shows that using these technologies can lead to higher product quality, less waste, and greater efficiency. Upon analyzing the data presented in Table 6, it is clear that the long-term benefits in terms of cost reduction and enhanced efficiency may surpass the initial expenses linked to the implementation of AI and ML. As firms further investigate the possibilities of new technologies, it is crucial to thoroughly evaluate the pros and cons and make well-informed choices that are in line with their strategic objectives and priorities.

## 7. Challenges and limitations

In order to fully realize the potential of AI vision and machine learning in the food processing sector, a number of obstacles must be overcome (Fig. 5). A primary obstacle is the substantial initial financial investment required to put these most recent innovations into practice. Purchasing and integrating advanced AI systems can be expensive, particularly for small and medium-sized businesses. The potential for cyberattacks and data leaks is still another significant worry. Food processing facilities are more susceptible to cyberattacks that could steal sensitive data and cause operational disruptions as they become more digitalized and networked. Strong cybersecurity defenses are essential to preventing such threats.

To maintain quality control and predictive maintenance standards, AI and machine learning systems must have high accuracy and reliability. While AI systems are often more accurate than traditional approaches, there are still challenges in maintaining consistent performance and reducing false positives and negatives. Issues with scalability and compatibility may potentially prevent AI technology from being widely used. It can be challenging to integrate AI with present systems, and it's critical to make sure they can grow effectively to accommodate rising production demands.

Another drawback is that managing and running AI-driven systems requires a trained team. Despite the time and expense needed, it is imperative to invest in training staff members to handle the latest

**Table 2b**
Government initiatives in the Indian subcontinent [as of June 2024].

| Implementing level | Objective | Reference |
|---|---|---|
| Ministry of Food Processing Industries (MoFPI) | The 2024 budget allocation aims to develop modern infrastructure for efficient supply chain management, provide financial assistance for food processing and preservation capacities, and encourage public-private partnerships in the food processing sector. | (Singh, 2024) https://mofpi.gov.in/Schemes/about-pmksy-scheme |
| Ministry of Food Processing Industries (MoFPI) | The PMFME scheme, with a budget of Rs. 100 billion ($1.3 billion) started in 2020, to upgrade 200 thousand food processing firms, must ensure government support is tied to easy-to-measure performance metrics, such as credit-linked subsidies, to prevent perverse incentives and ensure effective use of public funds. | (Gupta et al., 2023) https://pmfme.mofpi.gov.in/pmfme/newsletters/enews2.html |
| Ministry of Food Processing Industries (MoFPI) | The Mega Food Park Scheme launched in 2008, funded by the Indian government, aims to enhance the food processing industry with modern infrastructure, establish a robust supply chain, and reduce farm food wastage by linking farmers to the market. | (Singh and Sinha, 2018) https://mofpi.gov.in/Schemes/mega-food-parks |
| National level | The eNAM is an online trading platform, launched in April 2016 by PM of India, that facilitates transparent and efficient trade of over 90 agricultural commodities, enabling better price discovery and streamlined marketing for farmers, traders, and buyers. | (Prasad & Rao, 2019) |
| National level | AGRI-UDAAN (launched in 2015), is a 6-month accelerator program by a-IDEA, NAARM, supported by India's Department of Science & Technology, aimed at scaling up start-ups through mentoring, networking, and investor pitching. | (Kumar et al., 2020) |
| State level | The Government of Karnataka has signed an MoU with Microsoft in Oct, 2017 to empower smallholder farmers with AI-based solutions, using cloud technologies, machine learning, and advanced analytics to develop a multivariate agricultural commodity price forecasting model. | (Mahapatra & Singh, 2021) |
| National level | The Integrated Cold Chain and Value Addition Infrastructure Scheme, launched in 2008 by MoFPI, promotes seamless cold chain facilities from farm to consumer to reduce losses and improve efficiency in the collection, storage, transportation, and minimal processing of perishable products. | (Singh et al., 2018) |
| Ministry of Food Processing Industries (MoFPI) | The PMFME Scheme adopts the One District One Product (ODOP) approach in 2020 to develop value chains and infrastructure for perishable agro-produce, cereal-based, or traditional food products. It supports existing micro units and new units for ODOP products, with 75 Incubation Centres approved across 25 States/UTs. | (Anand et al., 2023) |
| National level | The Production Linked Incentive Scheme for Food Processing Industry (PLISFPI), approved in March 2021 as a part of the Aatmanirbhar Bharat Abhiyan, aims to create global food manufacturing champions, promote Indian food brands, increase off-farm employment, and ensure better prices for farmers from 2021 to 22–2026–27 with an outlay of Rs. 10,900 crores. | (Wandhe, 2024) |
| Ministry of Food Processing Industries (MoFPI) | The Scheme for Agro Processing Clusters, implemented since 2017, requires a minimum of 5 food processing units with a minimum investment of Rs. 25 crores per cluster. | https://mofpi.gov.in/Schemes/agro-processing-cluster |
| Ministry of Food Processing Industries (MoFPI) | The Scheme, launched in 2018, aims to integrate the processed food industry by linking farmers to processors and markets, with a revised allocation of Rs. 187.50 crore by 2019–20. | https://www.mofpi.gov.in/Schemes/scheme-creation-backward-and-forward-linkages |

technologies. It's also critical to address safety concerns and ethical dilemmas. The use of AI in the food processing sector raises questions about the moral implications of automation, including the potential displacement of labor and the transparency of AI decision-making processes. In order to gain the public's confidence and acceptance, AI systems must adhere to safety regulations and moral principles. While AI vision and machine learning have a lot to offer the food processing industry, their successful implementation and long-term sustainability depend on addressing issues like cybersecurity risks, cost, accuracy, scalability, workforce skills, and ethical considerations.

## 8. Future directions and innovations

By improving predictive maintenance, blockchain integration,

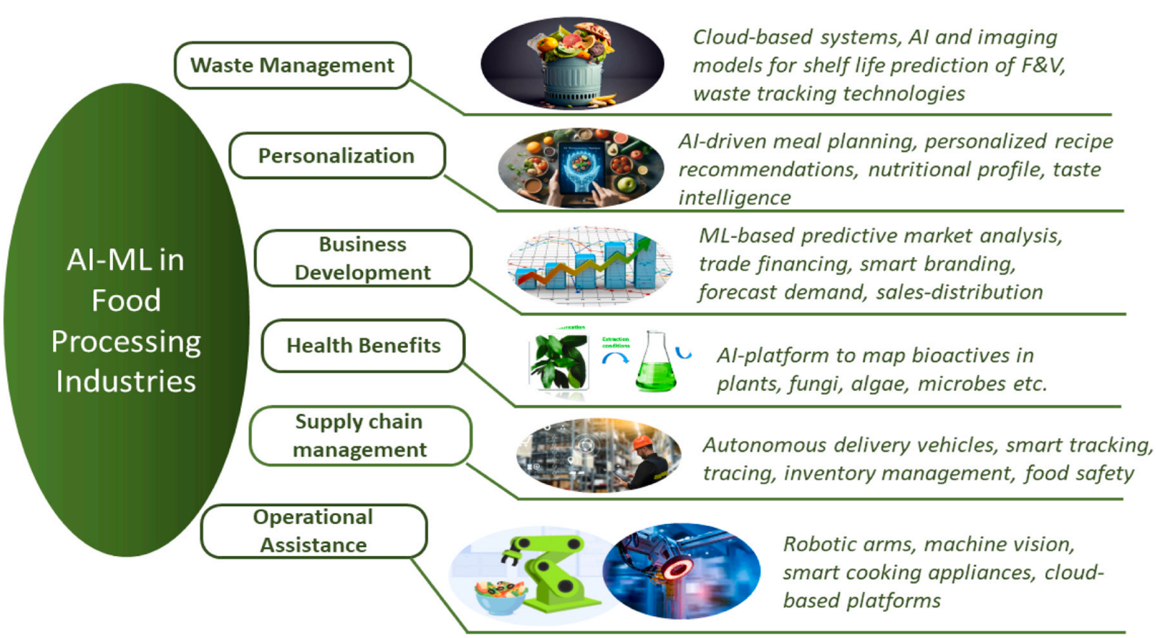


**Fig. 4.** Major applications of AI-ML in Food Processing Industries.

**Table 3**
Major companies employing AI Vision and ML in food processing and their broad areas of work.

| Company | Operating Region/ Country | Broad Area of Work | Revenue/ valuation/ funding | Lead investors/ major funding sourcess | Website |
|---|---|---|---|---|---|
| Afresh | San Francisco, California, USA (founded in 2017) | Their flagship product Fresh Operating system, uses AI-powered recommendations for inventory management, forecast demand, place accurate order for fresh produce, meat, seafood, other perishables. | $21 million (revenue) $148 million (funding) | Maersk Growth, High Sage, Innovation Endeavours, Walter Robb, Bright Pixel Capital etc. | https://www.afresh.com/ |
| Bear Robotics | Seoul, South Korea (founded in May, 2017) | Their' flagship product, Servi, an AI-powered robot assists restaurant staff with tasks like delivering food, bussing tables, and enhancing customer service. Their Robots-as-a-Service (RaaS) model provides these robots on a subscription basis. | $33.3 M | LG Electronics | https://www.bearrobotics.ai/ |
| Bizom | Bengaluru, India (founded in 2008) | Provides retail intelligence platform for CPG, FMCG brands and B2B retailers to digitize end-to-end sales and distribution using a proprietary analytics engine with AI & ML to deliver custom reports, alerts & actionable insights – also offers auto-replenishment system, business intelligence, smart merchandising, trade financing, trade promotion management. | $7.4 million (Revenue) $7.18 million (funding) | Pavestone, IndiaMART etc. | https://bizom.com/ |
| Blendid | Sunnyvale, California, US (founded in Nov, 2015) | Blendid is a fully autonomous robotic kiosk designed to prepare and serve a variety of blends. It features a robotic arm, extensive ingredient storage in freezers and refrigerators, a blending station, cup dispensers, and a water system. | $1.8 million (Revenue) $20 million (funding) | Green Cow Venture Capital, NBK INNOVATION XIII, Think+ etc. | https://www.blendid.com/ |
| Brightseed | South San-Francisco, California, US (founded in 2018) | Uses AI platform Forager, to discover and map bioactive compounds in plants, microbe and fungi for human health benefits. | under $20 million | AgFunder, Germin8, 3 C Capital, McKenny-McFarlane Capital etc. | https://www.brightseedbio.com/ |
| Crown Digital | Headquartered in Singapore (founded in 2018) | Manufactures automatic coffee-making robot that takes customer order via an app and provides real-time notification when the coffee is ready. | $3.8 million (Revenue) $6.68 million (funding) | Heliconia Capital, ParticleX etc. | https://www.crowndigital.io/ |

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**Table 3** (*continued*)

| Company | Operating Region/ Country | Broad Area of Work | Revenue/ valuation/ funding | Lead investors/ major funding sourcess | Website |
|---|---|---|---|---|---|
| Dexai Robotics | Boston, MA, USA (founded in 2018) | Develops robotic arm for automating activities in commercial kitchens and the food industry. | $4.5 million (Revenue) $15.5 million (funding) | 10X capital, AlleyCorp, etc. | https://www.dexai.com/ |
| ElasticRun | Headquartered in Pune, India (founded in 2016) | B2B platform connecting FMCG and grocery brands to rural markets using AI engines for supply aggregation and ML engines for store-level demand predictions. | $5.69 Billion (Revenue) $461 million (funding) | SoftBank Vision Fund, SoftBank Group and Goldman Sachs | https://www.elastic.run/ |
| FoodDocs | Tallinn, Estonia, Europe (founded in 2017) | Features AI-powered food safety apps that helps in meeting HACCP, PCP, ISO 22000, and BRC standards quickly and also provide Customize monitoring and food traceability. | $150100 (Revenue) $ 3.5 million (funding) | Bonnier Capital, Forward VC, Spring Capital, Ajujaht, EstBAN | https://www.fooddocs.com/ |
| Gastrograph | New York, US (founded in 2010) | Provides AI-based flavor profiling analytics solutions for food manufacturers by collecting data about food flavors, aromas, and textures. | $2.5 million (Revenue) $6 million (funding) | BASF, Meta Materials, Plug and Play Tech Centre etc. | https://www.gastrograph.com/ |
| Hungryroot | New York, USA (founded in 2015) | Uses AI to create personalized grocery lists and recipes based on individual dietary preferences, health goals, and cooking habits. | $333 million with a profit of over $9 million in 2023 | L Catterton and Lightspeed Venture Partners, Crosslink Capitals | https://www.hungryroot.com/ |
| Impossible Foods | Redwood City, California, USA (founded in 2011) | Aims to reduce environmental impact of meat production by offering nutritious alternatives with plant-based soy and heme proteins. | $256.5 million | Mirae Asset | https://impossiblefoods.com/ |
| Innit | Palo Alto, California (founded in 2013) | AI-driven meal planning, personalized recipe recommendations, food tracking and nutritional data features. | $16.0 million | Thomvest Ventures, Fairhaven capital, S2G Venture | https://www.innit.com/ |
| Miso Robotics | Pasadena, California, USA (founded in 2016) | Provides robotic technology for assisting chefs with a cloud-based learning platform that has the capability to automatically identify and locate food and tools through 3D, thermal, and regular vision. It also offers an autonomous robotic kitchen assistant that can learn from its surroundings and acquire new skills over | $3.0 million (Revenue) $15 million (funding) | Wavemaker Partners, Ecolab and Yosef Hertz | https://misorobotics.com/ |

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**Table 3** (*continued*)

| Company | Operating Region/ Country | Broad Area of Work | Revenue/ valuation/ funding | Lead investors/ major funding sourcess | Website |
|---|---|---|---|---|---|
| | | time and can help in assistance with activities such as grilling, frying, and prepping ingredients. | | | |
| NotCo | Santiago, Chile (founded in 2015) | Uses AI platform Giuseppe, to analyzes the molecular structure of animal products to find plant ingredients that mimic their taste, texture, and functionality. | $235 million with a valuation of $1.5 billion | Princeville Capital | NotCo https://notco.com/ |
| Nuro | Mountain View, California, US (founded in 2016) | Provides AI-enabled electric autonomous vehicles for deliveries that are equipped with cameras, Lidar, short and long-range radar, and ultrasonic sensors. | $438.8 million (revenue) $2.13 billion (funding) | T. Rowe Price, Greylock, Disney Accelerator | nuro https://www.nuro.ai/ |
| Ocado Group | Hatfield, England, UK (founded in 2000) | Offers platform for operating online grocery businesses with software systems including robotics, AI solutions for web-shops, voice ordering etc. | £ 2.8 billion (Revenue) | Lingotto Investment Management LLP, Baillie Gifford & Co., The Kroger Co. | ocado® GROUP https://www.ocadogroup.com/ |
| OneThird | Enschede, Netherlands (founded in 2019) | Provides food waste prevention solutions for offline businesses using a cloud-based system, scanners, AI and imaging models to predict the shelf life of fruits and vegetables and perform trend analysis. | $1.8 million (Revenue) $6.21 million (funding) | Pymwymic, Halma Ventures Ltd., SHIFT Invest, Oost NL | Onethird FOOD WASTE PREVENTION https://onethird.io/ |
| Paytronix | Newtown, MA, USA(founded in 2001) | Offers cloud-based customer experience management solutions for restaurants and convenience stores; with customer insights using AI and ML. | Generates 25–30 % of all orders digitally. $300 billion (Revenue) $75 million (funding) | Great Hill, SVB | PAYTRONIX https://www.paytronix.com/ |
| Rotimatic | Singapore (founded in 2008) | Develops fully automatic flatbread-making robot. The robot comes with built-in artificial intelligence and IoT capabilities with a 32-bit microprocessor that harmoniously orchestrates 10 motors and 15 sensors. Can prepare up to 20 loaves of bread in one go and mirrors human judgment to adjust the proportion of flour and water in real-time. | $21.2 million (Revenue) $48.5 million (funding) | Credence Partners, EDBI, NSI ventures, and Robert Bosch Venture | https://in.rotimatic.com/ |

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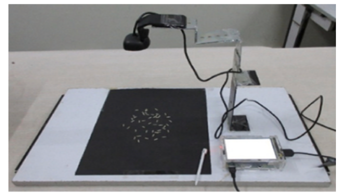
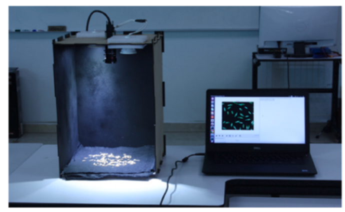
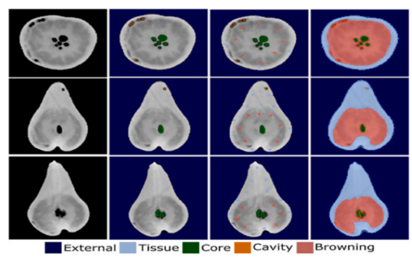
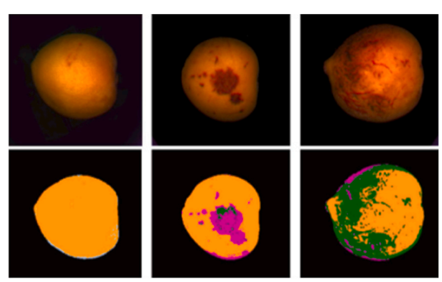
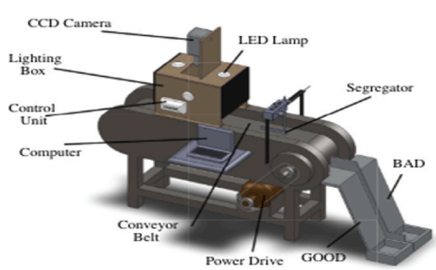
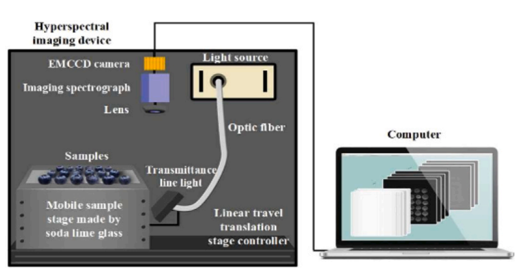
Table 3 (*continued*)

| Company | Operating Region/ Country | Broad Area of Work | Revenue/ valuation/ funding | Lead investors/ major funding sourcess | Website |
|---|---|---|---|---|---|
| SnapCalorie | New York, USA (founded in 2019) | Offers platform for food detection solutions using AI. The customers can take a picture of the meal and it provides nutritional information including calories, fat, carbs, and protein. | $1.5 million (revenue) $2.12 million (funding) | Y Combinator, Index Ventures and Accel | https://www.snapcalorie.com/ |
| Spoonshot | Bengalore, India (founded in 2015) | Offers platform to predict consumer tastes and food trends for restaurants and foodservice companies using food science and machine learning. Provides - taste intelligence, personalized recommendation, and guest intelligence. | $123000 (Revenue) $1.8 million (funding) | SRI Capital | https://spoonshot.com/ |
| StandardAI | San Francisco, CA (founded in 2017) | AI-enabled automated checkout software solutions for offline retailers using cameras that tracks the movement of items with 3D analysis. | $29.4 million (Revenue) $231 million (funding) | Y Combinator, SoftBank, CRV and EQT Ventures | https://standard.ai/ |
| The whole Truth Foods | Mumbai, India (founded in 2019) | Multi-category snacks brand using AI-platform to allow users regarding fitness inquiry. | $5.78 million (Revenue) $22.4 million (funding) | Peak XV Partners | https://thewholetruthfoods.com/ |
| Upliance.ai | Bengaluru, Karnataka, India (founded in 2021) | Offers online platform for AI-integrated smart cooking appliances. Its features include temperature control system, a touch screen with the in-built recipe, integrated weighing scale, etc and performs functions like self-cleaning, stirring, blending, mixing, grinding, and chopping. | $17.1 million (valuation) $5.72 million (funding) | Khosla Ventures, Draper Associates and Rainmatter | https://upliance.ai/ |
| Vivid Robotics (Picnic) | Dubai (founded in 2022) | Develops automated robotic pizza system for food preparation as per customized ingredients and recipes | $7.37 million (funding) | Vulcan Capital, Draper Associates and Wisemont Capital | https://www.picnicworks.com/ |
| Winnow | London, UK (founded in 2013) | Provides platform for food waste tracking and management solutions; equipped with smart waste metering technology. | $24 million (Revenue) $34.1 million (funding) | ABC Impact, PRETB, ArcTern Ventures. | https://www.winnowsolutions.com/ |

*CPG: Consumer packaged goods; FMCG: Fast-moving consumer good; LiDAR: Light Detection and Ranging; HACCP: Hazard Analysis and Critical Control Point; PCP: Preventive Control Plan; ISO 22000: A food safety management system (FSMS) developed by International Organization for Standardization (ISO); BRC: British Retail Consortium.

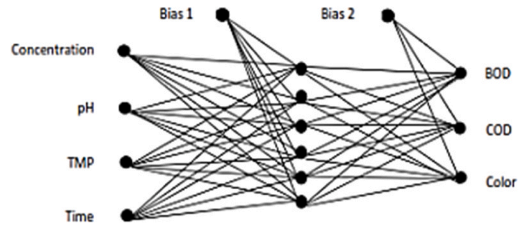
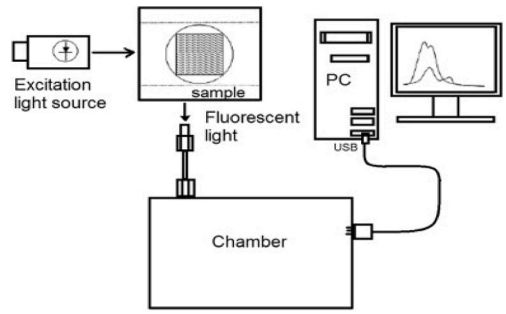
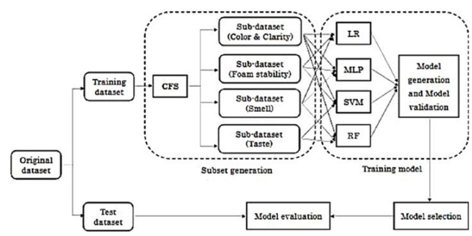
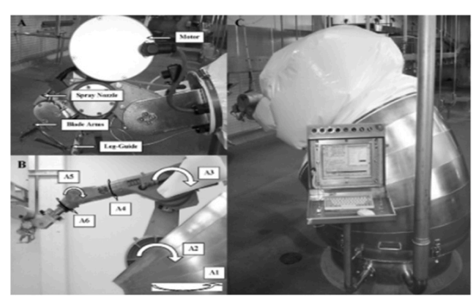
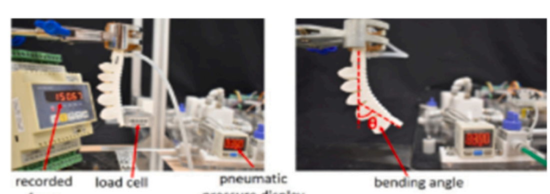
energy management, creative food formulation, virtual and augmented reality integration, and 3D printing technology, emerging technologies in artificial intelligence and machine learning are revolutionizing the food processing sector (Fig. 6). Through real-time monitoring and proactive interventions, predictive maintenance solutions reduce equipment downtime and increase machinery longevity. Blockchain integration transforms the transparency and traceability of the supply chain, guaranteeing food safety and legal compliance (Puthenveettil & Sappati, 2024). Within food processing facilities, AI-powered energy management solutions maximize resource usage, cut waste, and support sustainable practices. AI-driven research makes it easier to create novel food formulations that result in food products that are more sustainable

**Table 4**
Summary of AI vision and ML in food processing with notable features and performance metrics.

| Technology used | Application/ Description | Performance Metrics | Block Diagram/ Mechanical setup/Analysis | References |
|---|---|---|---|---|
| Image Processing, Support vector machine (SVM), ML | Non-destructive assessment of rice quality for commercial valuation based on extraction of geometrical features like elongation, eccentricity and chalk ratio. | 93 % |  | (Mittal et al., 2019) |
| Vision and ML algorithms, SVM-RBF, Python interface on a MATLAB engine | Assessment of the quality of malting barley grain by extracting color, shape and extraction features | 85–100 % |  | (Ramirez-Paredes & Hernandez-Belmonte, 2020) |
| X-ray, CT scan, DL | Non-destructive inspection of internal disorders in pear fruit like internal browning and cavity formation during CA storage | 92.2–99.4 % |  | (Van de Looverbosch et al., 2021) |
| ML algorithms, LF NMR, DLNN, PCA | Detection of internal fungal infection and pest invasion in dried longan fruits without peeling based on LF NMR characterized bound water in the fruits. | 89 % | | (Fu et al., 2022) |
| ML models (PLS, RF and XGBoost), VIS and NIR hyperspectral imaging | Differentiation of internal physiological and external mechanical defects in loquat fruit. | > 95 % |  | (Munera et al., 2021) |
| Machine vision (CCD camera, microcontroller, sensors, | Online sorting of tomatoes based on shape (oblong and circular), size (small and large), maturity (color), and defects. | 84.4–94.5 % |  | (Arjenaki et al., 2013) |
| DL, CNN, ResNet, ML algorithms (SMO, LR, RF, Bagging and MLP) | Detection of internal mechanical damage of blueberries using hyperspectral transmittance data | 98 % |  | (Wang et al., 2018) |

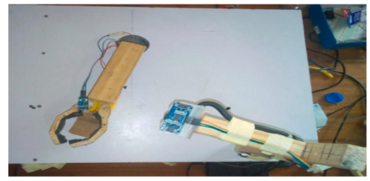
(continued on next page)

**Table 4** (*continued*)

| Technology used | Application/ Description | Performance Metrics | Block Diagram/ Mechanical setup/Analysis | References |
|---|---|---|---|---|
| ANN, Membrane based RO, MATLAB | Optimization of Palm Oil Mill Effluent (POME) treatment using RO membrane based on BOD, COD, TSS, color, pH and turbidity | 88.14–96.12 % |  | (Said et al., 2018) |
| LED, CCD array sensor, ANN | classification vegetable oils like: Canola, Sunflower, Corn and Soybean from different manufacturers (that cannot be done by only observing the color, odor or taste) | 72 % |  | (da Silva et al., 2015) |
| ML models (LR, SVM, RF, MLP), CFS technique | Sensory evaluation of Saigon Beer based on the correlation between physicochemical components ($CO_2$ content, ethanol, acidity, soluble content, r Diacetyl and Other Diketone Substances) and sensory factors (clarity and color, foam stability, flavor and taste). | – |  | (Lu et al., 2020) |
| HPLC, LC–MS/MS, Electronic tongue | Analysis and evaluation of EUC in mushroom extracts. | > 95 % | | (Phat et al., 2016) |
| AI, WSN, ANN | Designing green and low-carbon food packaging | 97.6 % | | (Dai, 2023) |
| Robotic system, neural network-based fault detection scheme | Automation of meat-processing task with continuous optimization of the robot cut-path ensuring g improved aseptic conditions, reliability, repeatability and quality of a butcher's operation irrespective of seasonal carcass variations. | 98 % |  | (Hurd et al., 2005) |
| Versatile soft robotic gripper system, vision and tactile sensors, 3D printing | Potential application in food and grocery supply chain with 3D printed fingers reconfigured for different poses such as scoop, pinch, and claw to recognize and safely grip foods like uncooked tofu, broccoli floret, sausages etc. | 93.3 % |  | (Low et al., 2021) |
| Pneumatic Robot arm, Chromatic sensor | Sorting of objects from a set according to their color. It can sense damaged object and There is no angle problem, even better than RFID. | – |  | (Sughashini et al., 2021) |

(*continued on next page*)

**Table 4** (*continued*)

| Technology used | Application/ Description | Performance Metrics | Block Diagram/ Mechanical setup/Analysis | References |
|---|---|---|---|---|
| Automated liquid filling system, robotic arm conveyor, Arduino Mega 2560, IR proximity sensors, ultrasonic sensor | Automatically fill number of container(s) with a liquid to a desired level by sensing the presence of bottle and the level of liquid. | – |  | (Abubakar et al., 2022) |

*CA: Controlled Atmosphere storage; LF NMR: low field nuclear magnetic resonance; DLNN: Deep learning Neural Network; PCA: Principal Component Analysis; VIS: visible; NIR: near infrared; CNN: Convolutional Neural Network; ResNet: Residual Network (improved version: ResNeXt); SMO: Sequential Minimal Optimization; LR: Linear Regression; RF: Random Forest; MLP: Multilayer Perceptron; ANN: Artificial Neural Network; RO: reverse osmosis; BOD: biological oxygen demand; COD: chemical oxygen demand; TSS: Total suspended solids; LED: Light emitting diode, CCD: charge-coupled device; CFS: Correlation-based feature selection; HPLC: High performance liquid chromatography; LC: Liquid chromatography; MS: mass spectroscopy; EUC: Equivalent umami concentration; WSN: wireless sensor network.

and healthier. Integrating virtual and augmented reality improves staff training programs, increasing operational effectiveness and offering immersive experiences for process optimization and quality control. Implementing AI into innovative food processing techniques like non-thermal processing (Sarkar et al., 2024); plant-based processing (Mullaiselvan et al., 2024; Saha et al., 2024); bioprocessing techniques (Dubey & Tripathy, 2024); 3D food printing technology (Kadival et al., 2023; Padhiary et al., 2024; Pradhan et al., 2024), nanotechnology (Padhiary et al., 2025); will create a new realm for specialized food formulations and personalized dietary preferences.

These developments are accelerating the food industry's progress toward a future characterized by greater sustainability, innovation, and efficiency. A number of areas call for additional study and development in order to fully realize the potential of AI and machine learning in the food processing industry. Accuracy and dependability will be increased by enhancing predictive maintenance through the integration of complex AI algorithms and IoT devices. Research on AI-powered energy management systems may lead to lower environmental impact and more effective use of resources. AI and machine learning-based food formulation innovations have the potential to provide food items that are more sustainable and healthier. To fully realize the advantages of virtual and augmented reality integration in training, quality assurance, and process optimization, more work needs to be done. Diverse studies can open up new avenues for efficiency and innovation in the food processing sector

**Table 5**
Patent title, code, and application of AI-ML in food processing.

| Title | Patent code | Product/ Design Algorithm | Application(s) |
|---|---|---|---|
| Automatic vision guided intelligent fruits and vegetables processing system and method | US11531317B2 | Intelligent guided system, 2nd generation strawberry decalyxing system (AVID2) using CNN | Make decisions whether a strawberry is to be cut or rejected, and compute a multi-point cutline curvature to be cut along by rapid robotic cutting tool. |
| Automated food freshness detection using feature deep learning | AU2020100953A4 | Hybrid architecture using DL algorithm (YOLOV3, Google Net inception V4, SoftMax) | Food recognition and quality detection of vegetables, fruits, meat, aquatic products, nutrients estimation |
| Food testing system and program | JP2019211288A | Food testing system using DL and multivariate analysis | Image acquisition, feature extraction, automatic sorting of non-defective or defective food. |
| Delivery of food items by aerial or ground drones to and from delivery vehicle | US11816624B2 | Autonomous or semi-autonomous drones | Deliver food items (can carry multiple payloads for multiple locations), have environmentally controlled storage. |
| Automated crab meat picking method | US11882843B1 | Image processing system | Crab meat image being processed, then cut in accordance with the appropriate cutting lines developed |
| Automated preparation and dispensation of food and beverage products | US11657669B2 | Cloud based management system, automated food production system using ML algorithm | Robotic arm for performing food preparation tasks, with different recipes stored in cloud server. |
| Artificial intelligence-based food flavour sensory evaluation system and establishment method thereof | WO2019114052A1 | AI-based food flavour sensory evaluation system | Utilizing GC-MS for obtaining flavour feature original data of foods; GC-O for obtaining artificial sensory evaluation data of food flavours – more reliable |
| Meal delivery method, system, medium and intelligent device based on meal delivery robot | CN110710852B | Medium and intelligent robot using laser radar | Face recognition for taking orders, efficient meal delivery by robots. |
| Systems and methods to mimic target food items using artificial intelligence | US20220005376A1 | Prediction model based on RNN | System learns from open source and proprietary databases, determine a recipe to mimic the target food item |
| Robotic kitchen hub systems and methods for minimanipulation library adjustments and calibrations of multi-functional robotic platforms for commercial and residential enviornments with artificial intelligence and machine learning | US20210387350A1 | AI-ML based robotic kitchen hub | Robotic platform includes a robotic kitchen for calibration, food dish preparation by robot, operational assistance or robot-user collaboration. |
| Blood and saliva biomarker optimized food consumption and delivery with artificial intelligence | US20180293638A1 | Cloud based computer implemented method using recursive techniques and neural networks | Optimize food and beverage nutrient efficiency into the user's blood chemistry. |

*CNN: Convolutional Neural Network; DL: Deep Learning; GC-MS: Gas Chromatography- Mass Spectroscopy; GC-O: Gas Chromatography- olfactometry; RNN: Recurrent Neural Network

**Table 6**
Comparative performance and cost economics of traditional methods versus AI and ML-enhanced methods in food processing.

| Aspect | Traditional Methods | AI and ML-Enhanced Methods | Comparison | Example | References |
|---|---|---|---|---|---|
| Initial Investment Costs | $50,000 - $100,000 for basic equipment, manual labor | $200,000 - $500,000 for advanced hardware, software, integration | AI and ML require higher upfront costs for technology and training. | A food processing plant investing $350,000 in AI Vision systems for automated quality control | (Nicholas-Okpara et al., 2021) |
| Operational Costs | $150,000 annually (manual labor, maintenance) | $100,000 annually (maintenance, updates, data processing) | AI and ML reduce manual labor costs but involve ongoing maintenance and updates. | Annual maintenance and updates for AI systems costing $100,000 | (Knox, 2023) |
| Quality Control | Manual inspection, 90 % accuracy, prone to human error | Automated, 99 % accuracy, consistent | AI and ML offer superior accuracy and reduce error rates. | AI Vision system achieving 99 % accuracy in detecting defects | (Azamfirei et al., 2023) |
| Predictive Maintenance | Reactive, scheduled maintenance, 10 % downtime | Proactive, real-time monitoring, 2 % downtime | AI and ML minimize unexpected downtimes and extend equipment lifespan. | Predictive maintenance reducing downtime to 2 %, saving $50,000 annually | (Appiah et al., 2022) |
| Process Automation | Limited automation, labor-intensive | High automation, efficient task handling | AI and ML increase efficiency and throughput through advanced automation. | Robotics guided by AI reducing labor costs by 30 % | (Tschang & Almirall, 2021) |
| Supply Chain Optimization | Manual inventory management, 85 % efficiency | AI-driven analytics, 95 % efficiency | AI and ML enhance inventory accuracy and reduce logistical costs. | AI optimizing supply chain, reducing logistics costs by 20 % | (Ferreira & Reis, 2023) |
| Personalization in Production | Limited customization, mass production | High customization, tailored products | AI and ML enable personalized offerings, meeting niche market demands. | Personalized nutrition products increasing sales by 15 % | (Amoako et al., 2021) |
| Long-term Financial Benefits | Incremental improvements | Significant cost savings and revenue growth | AI and ML drive long-term gains through efficiency and new revenue streams. | AI systems leading to annual savings of $200,000 | (Davenport & Mittal, 2023) |
| Return on Investment (ROI) | Slower ROI due to inefficiencies | Faster ROI from efficiency gains and cost reductions | AI and ML can provide a quicker and higher ROI despite higher initial costs. | ROI achieved in 3 years, with annual cost savings of $150,000 | (George, 2024) |

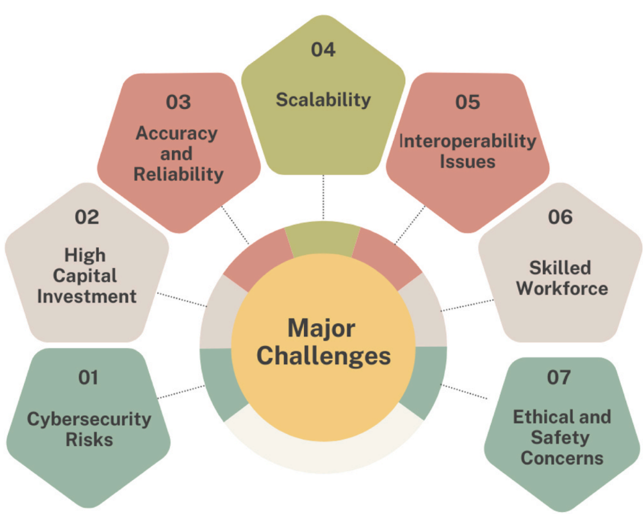



**Fig. 5.** Major Challenges in implementing AI-ML in food processing.

by fusing AI with other innovative technologies.

## 9. Conclusion

The food processing industry has seen notable and revolutionary advancements as a result of the integration of AI technologies for machine learning and image identification. The present study highlights how important artificial intelligence is to enhancing robotics, supply chain optimization, robotic quality assurance, predictive maintenance, and customized production. Compared to the 80–90 % accuracy of conventional manual inspections, AI visual recognition systems have redefined quality assessments, ensuring consistent product quality and safety. They have automated quality assessments with nearly 99 % accuracy. By cutting downtime from 10 % to only 2 %, predictive maintenance has improved total production efficiency. Artificial intelligence has improved and automated robotics, resulting in 30 % lower labor costs and more effective procedures. Furthermore, improved inventory and logistics management have resulted from supply chain optimization, with up to 95 % efficiency, despite the hefty initial capital investments needed. AI-enabled Personalized items can increase sales by 15 %, but traditional approaches only allow for limited customization. In contrast, AI and ML provide significant personalization.

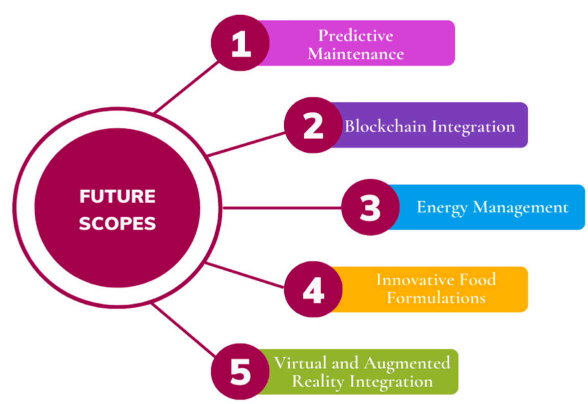



**Fig. 6.** Future scopes of AI-ML in food processing.

AI in food processing has an abundance of prospective applications and possible advantages. Startups and Government organizations are keen on developing systems based on these applications, with the majority of these works being in the development/ prototyping/ production/ testing stages. Some of the promising works include the US-AI Center of Excellence, China's NDRC initiatives on food industries, the UK's Agritech Strategy, India's PMFME Scheme and Mega Food Park Scheme by MOFPI; some Indian startups viz., Bizom's AI platforms, Upliance's smart cooking devices, Spoonshot's sensory prediction platform, and so on. The trend suggests future advancements in AI-powered quality control, individualized nutrition, sustainable projects, and seamless supply chain integration. The advancement of AI in food processing necessitates the resolution of issues pertaining to labor readiness, ethical considerations, data protection, and integration challenges.

Industry executives, legislators, and engineers must work together skillfully to design an AI-powered future that promotes ethical responsibility in the food processing sector in addition to technological growth.

## Author contributions

Debapam Saha contributed in Writing-original draft, Visualization, Resources, Methodology, Investigation, Conceptualization. Mrutyunjay Padhiary contributed in Writing and reviewing the original draft & editing, Supervision, Software, Resources. Naveen Chandrakar had done Data curation, Formal analysis, Resources.

## Statements and declarations

None.

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No funding was received for conducting this study.

## CRediT authorship contribution statement

**Padhiary Mrutyunjay:** Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Conceptualization. **Saha Debapam:** Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Conceptualization. **Chandrakar Naveen:** Resources, Formal analysis, Data curation.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

This is a review article and the data has been collected from literature, Internet Articles, and Government official websites. All the citations are included in the reference section.

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